

CASEORA

AI-Powered Advocate Case Management System — Developer Blueprint

"One Case. One Timeline. Complete Control."

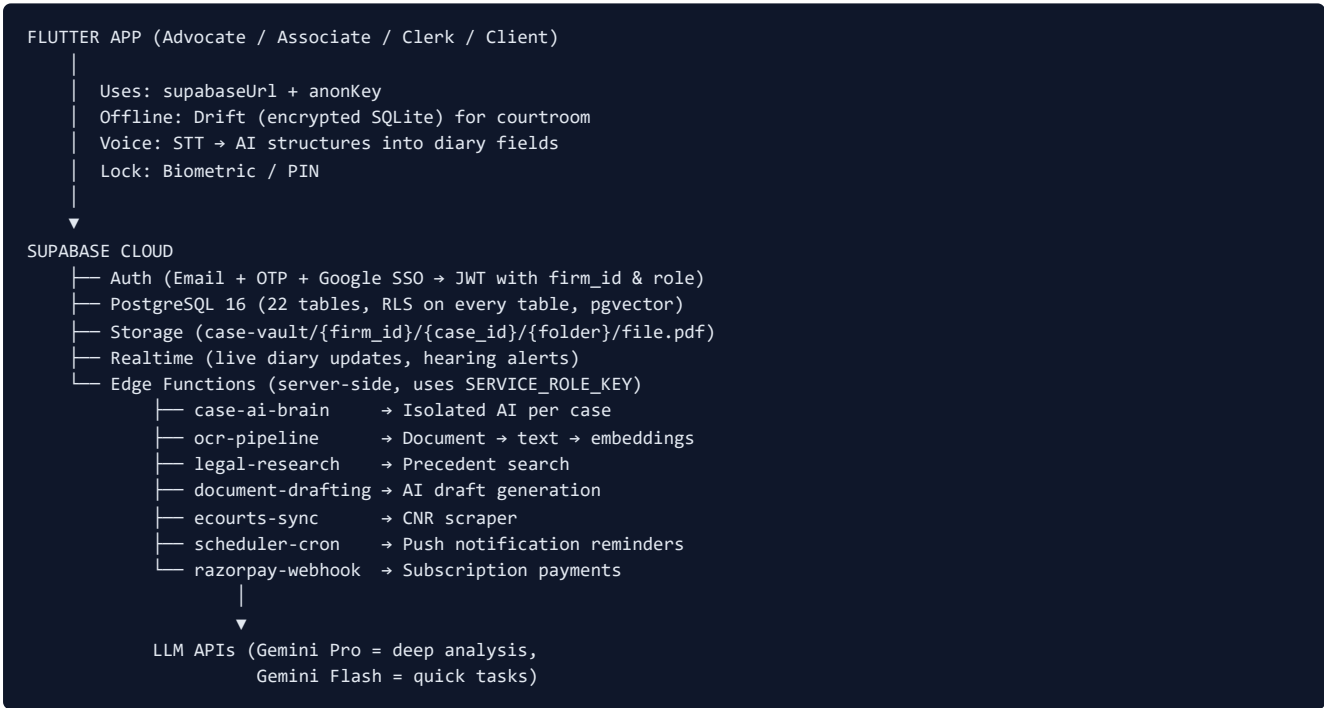
1. Keys & Credentials You Need

Key	Get From	Use Where
SUPABASE_URL	Supabase → Settings → API → Project URL	Flutter app init
SUPABASE_ANON_KEY	Supabase → Settings → API → anon public	Flutter app init
SUPABASE_SERVICE_ROLE_KEY	Supabase → Settings → API → service_role secret	⚠ Edge Functions ONLY
GOOGLE_AI_API_KEY	aistudio.google.com	Edge Function secret
OPENAI_API_KEY	platform.openai.com (for embeddings)	Edge Function secret
RAZORPAY_KEY_ID + SECRET	dashboard.razorpay.com	Edge Function + Flutter
Firebase google-services.json	Firebase Console	Android app (push notifications)

⚠ **SERVICE_ROLE_KEY** bypasses ALL Row Level Security. NEVER put it in the Flutter app or GitHub repo. Only use inside Edge Functions (server-side).

Supabase Extensions to Enable: `uuid-oss` · `pgcrypto` · `vector` (pgvector)

2. System Architecture



Storage Path Pattern:

```
case-vault/{firm_id}/{case_id}/petition_plaint/document.pdf
case-vault/{firm_id}/{case_id}/exhibits/exh_p1.pdf
case-vault/{firm_id}/{case_id}/voice_memos/hearing_2024-09-25.m4a
```

3. Implementation Plan (6 Phases / 24 Weeks)

Phase	Weeks	What You Build
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1. Foundation	1–4	Flutter project setup · Supabase Auth (Email/OTP/Google) · Biometric lock · Advocate Profile · Firm creation · Team management with 2 free slots trigger · Premium upgrade paywall
2. Case Engine	5–8	Case registration (12 types) · Party details · Court/CNR/FIR details · Private Case Diary (what happened + court said + advocate submitted) · Voice-to-Diary (EN/MR/HI) · Hearing Calendar · Push notifications · Exemption management
3. Docs & Offline	9–12	Document Vault (14 folders per case) · Exhibit registry · Order management + AI summary · Case Timeline view · Case Disposal + Archive · Offline Court Mode (Drift SQLite + background sync)
4. AI Brain	13–17	OCR pipeline + embeddings (partitioned by case_id) · AI Case Summary · AI "What Happened?" · AI "What Next?" · Case Health Score (0-100%) · Ask CASEORA AI (single-case Q&A) · Legal Research + Precedent Finder · AI Drafting Assistant · Citation Extractor
5. Portal & Billing	18–21	Client Portal (read-only) · Secure messaging · Billing & expenses + invoice PDF · Legal Workflow Engine (step-by-step per case type) · Multilingual UI (English + Marathi + Hindi)
6. Launch	22–24	eCourts CNR sync (official data separated from private notes) · Security audit + RLS testing · Play Store + App Store deployment

4. Database Architecture (22 Tables)

Auth & Team

Table	Purpose	Key Columns
firms	Law firm / chambers (multi-tenant root)	id PK · admin_advocate_id FK→profiles · subscription_tier (free/premium/enterprise)
profiles	Users linked to Supabase Auth	id PK = auth.users.id · firm_id FK→firms · role (admin/advocate/associate/clerk/client) · enrollment_number · bar_council_state
team_members	Team slot assignments	firm_id FK · user_id FK · role · DB trigger blocks 3rd member on free plan

Clients & Cases

Table	Purpose	Key Columns
clients	Client records per firm	firm_id FK · client_name · phone_number · portal_access_enabled · portal_user_id FK→auth.users
cases	Core case table	firm_id FK · case_number · case_type · case_status · case_priority · client_id FK · advocate_client_role · lead_advocate_id FK · court_name · judge_name · cnr_number · fir_number · police_station · relevant_acts_sections · health_score (0-100) · is_disposed
case_parties	All parties in a case	case_id FK · party_name · party_designation · is_our_party · advocate_name (opposite counsel)

Hearings & Diary

Table	Purpose	Key Columns
case_hearings	Scheduled hearings	case_id FK · hearing_date · purpose · next_hearing_date · is_attended
case_diaries	Private advocate diary (Clerks/Clients BLOCKED)	case_id FK · what_happened · court_remarks · advocate_submissions · is_client_present · is_opposite_present · is_exemption_filed · court_decision · next_steps · audio_memo_url
exemption_applications	Client absence tracking	case_id FK · absence_reason · court_response · cost_imposed

Tasks & Documents

Table	Purpose	Key Columns
case_tasks	Deadlines & to-dos	case_id FK · title · due_date · priority · status · assigned_to FK→profiles
case_documents	Document vault	case_id FK · category (14 types) · storage_path · ocr_completed · extracted_text · ai_summary · is_client_accessible
case_exhibits	Marked exhibits	case_id FK · exhibit_number ("Exh. P-1") · document_id FK · witness_name · marking_date
case_orders	Court orders	case_id FK · order_date · order_type · ai_order_summary · compliance_required · compliance_deadline

AI Brain (Isolated Per Case)

Table	Purpose	Key Columns
case_ai_brain_state	AI analysis state (1-to-1 with case)	case_id PK+FK · facts_summary · what_happened_summary · what_next_actions (JSONB) · readiness_factors (JSONB)
case_document_embeddings	Vector chunks for semantic search	case_id FK (isolation key) · document_id FK · content_chunk · embedding VECTOR(1536) · HNSW index · Search always filters WHERE case_id = target
legal_precedents	Cached precedent library	citation (UNIQUE) · case_title · court_name · ratio_decidendi · embedding VECTOR(1536)

Timeline, Disposal & Billing

Table	Purpose	Key Columns
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case_timeline_events	Visual case stages	case_id FK · stage_name · event_date · is_completed
case_disposals	Final result (1-to-1 with case)	case_id FK (UNIQUE) · disposal_date · final_result (allowed/dismissed/acquitted/convicted/settled/withdrawn/etc.) · appeal_deadline
billings	Invoices	case_id FK · client_id FK · total_fee · paid_amount · pending_amount (auto-calculated)
case_expenses	Court fees, typing, travel	case_id FK · expense_name · amount · is_reimbursed
client_messages	Advocate ↔ Client chat	case_id FK · sender_id · recipient_id · message_text
audit_logs	Security trail	firm_id FK · user_id · case_id · action_type · metadata (JSONB)

5. RLS — Who Sees What

Data	Admin Advocate	Associate	Clerk	Client
All firm cases	✅ All	Assigned only	Read only	Own case only
Private Diary	✅ All	Assigned only	❌ BLOCKED	❌ BLOCKED
Documents	✅ All	Assigned only	Read only	Only if is_client_accessible = true
AI Brain	✅ All	Assigned only	❌	❌
Billing	✅ All	Read	Read	Own invoices
Audit Logs	Read (own firm)	❌	❌	❌

6. AI Isolation Rule (Most Important)

Every AI query:

1. Must include `case_id`
2. Fetches ONLY that case's data from DB
3. Searches ONLY that case's embeddings (`WHERE case_id = X`)
4. Builds fresh context (no persistent memory between queries)
5. After response → context is destroyed

Result: Case A's data can NEVER leak into Case B. Ever.

7. Flutter Project Structure

```
lib/
├── core/           → config, theme, constants, supabase_service, router
├── features/
│   ├── auth/       → login, profile setup
│   ├── dashboard/  → main dashboard
│   ├── cases/      → register, list, detail
│   ├── diary/      → private diary + voice-to-diary
│   ├── calendar/   → hearing calendar
│   ├── tasks/      → task management
│   ├── documents/  → vault, exhibits, orders
│   ├── ai_assistant/ → AI brain, research, drafting
│   ├── team/       → team management + 2 free slots
│   ├── billing/    → fees, expenses, invoices
│   └── client_portal/ → client read-only view
└── offline/       → Drift DB + sync service
```

Key Packages: `supabase_flutter` · `flutter_riverpod` · `drift` · `local_auth` · `flutter_secure_storage` · `table_calendar` · `file_picker` · `flutter_pdfview` · `google_fonts` · `intl` · `fl_chart`